

Relationship Between Age and Entrepreneurship: Examining Age, Cohort, and Period Effects for Three Countries

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Abstract

Entrepreneurship is an age-dependent phenomenon. We still lack clear evidence on how the age-entrepreneurship relationship is precisely shaped, how this shape changes over generations, as well as to what extent it differs between countries. By disentangling age effects from cohort and year effects, we investigate how entrepreneurship varies across the life cycle in Germany, the UK, and the US, using large harmonized individual-level panel data from 1992 to 2019. We show that there are common age profiles across cohorts, but only until age 60. Rather than a downward slope immediately after midlife, we observe a stable age-entrepreneurship relationship, followed by a delayed downward slope at higher ages. We also reveal several differences between the three countries in terms of the steepness and the precise peak of each slope. We further disclose generational differences (cohort effects) in the probability of being entrepreneurially active. The level of entrepreneurial activity appears to be higher among the youngest observable generation (Generation Y) in the US and Germany than among the Generation X, but not in the UK.

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JEL classification: L26, M13, I12

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1. Introduction

When in life are people most likely to start businesses? Research suggests an inverted u-shape, with entrepreneurship peaking in middle age (Kautonen et al., 2017; Lévesque and Minniti, 2006; Azoulay et al., 2020; Parker, 2009; Kautonen et al., 2014; Wickstrøm et al., 2022; Xiao and Wu, 2021; Andersson and Hammarstedt, 2010). But the precise contours of this life-cycle pattern are still unknown, because age effects are easily confounded with differences across cohorts and time periods. Whether entrepreneurship truly follows the same curve across generations remains an open question.

Moreover, entrepreneurial activities depend on the context individuals experience (Welter, 2011). This may affect the age–entrepreneurship relationship in two ways. First, entrepreneurial activity may vary by cohort, as formative experiences, or educational expansion shape skills, capital, and opportunities in different ways (Mannheim, 1952; Glen H. Elder, 1974; Schofer and Meyer, 2005; Thurik et al., 2013). Second, it may also vary across time periods, as macroeconomic conditions, external shocks, or policy measures directly affect entrepreneurial dynamics (Barlevy, 2007; Clementi and Palazzo, 2016; Konon et al., 2018). Building on core accounts of human capital, social networks (Becker, 1991; Aldrich and Zimmer, 1986a), and financial life-cycles (Modigliani and Brumberg, 1954; Ando and Modigliani, 1963), we develop and test a life-cycle framework that separates age, cohort, and time period influences on entrepreneurial activity.

Because age, time period, and cohort effects remain understudied, knowledge about the entrepreneurship across the life cycle is incomplete. Methodologically, much research is based on cross-sectional analysis or simplifying assumptions in longitudinal studies. As a result, changes in entrepreneurship across ages could reflect age effects, such as job-specific human capital, or cohort differences, such as education levels. More formally, age, period, and cohort are linearly dependent ($Age = Year - BirthYear$), making it impossible to estimate all three without

further assumptions.

Despite substantial scholarly interest (see the meta-analysis by [Zhao et al., 2021](#)), prior work has not disentangled age, period, and cohort effects, leaving estimated age profiles potentially biased. The implications of these distinctions are also substantial for practitioners and policymakers. For instance, addressing health differences, *i.e.* age effects, requires different measures than those needed to mitigate educational disparities, *i.e.* cohort effects. Rigorous identification of these effects benefits both theory and policy.

To close this gap, we analyze how entrepreneurship varies over the life cycle of individuals and which differences emerge across cohorts and calendar years. After conceptualizing how age, year, and cohort differences may shape the age-entrepreneurship relationship, we estimate age profiles of entrepreneurship¹ that are identified separately from cohort and year effects. By investigating cohort effects, we are able to reveal generational differences in entrepreneurial intentions, a topic hotly debated in societies. To examine the generalizability of our findings, we perform this analysis across three countries that are focal to the literature on entrepreneurship: Germany, the UK, and the US.

We use the Comparative Panel File (CPF v1.5), which harmonizes household surveys from multiple countries. We analyze Germany, the UK, and the US (1992–2019), combining the SOEP, BHPS/UKHLS, and PSID into 955,000 observations of 151,000 individuals.

In our empirical analysis, we follow [MaCurdy and Mroz \(1995\)](#) and [Fitzenberger et al. \(2021\)](#) to disentangle age, cohort, and year effects through functional form assumptions. Specifically, we test the separability assumption postulating that age and cohort effects do not interact. The latter assumption is fundamental: if the separability assumption holds, then the identified age profile is uniform across cohorts up to constant cohort shifts. Otherwise, any discussion of age profiles refers

¹The age profile of entrepreneurship is a curve depicting the share of entrepreneurs in the population for each age group.

to average age profiles and, hence, may mask substantial variation of age profiles across cohorts.

Our results yield novel insights into age, cohort, and year effects across the three countries. First, in all three countries, age profiles are uniform across cohorts between 25 and 60. Second, across countries, the age profiles between 25 and 60 resemble an inverted u-shape in the left tail that, however, transitions into a plateau in the right tail. Third, entrepreneurship declines only after 60, and the pace of exit varies by cohort. Fourth, while the slope of the age profiles is identical for all cohorts until age 60, there exist remarkable differences when investigating *level* shifts across cohorts. Most importantly, for the youngest observable cohorts in this analysis (Generation Y), we find higher levels of entrepreneurship in Germany and the US than for the Generation X, contradicting perceptions that younger cohorts lack entrepreneurial inclination (Parry and Urwin, 2011; Costanza et al., 2012). Yet, these conclusions do not apply to the UK, where levels of entrepreneurship indeed decline among the most recent cohort. Finally, across the three countries we observe contrasting macroeconomic effects on entrepreneurship confirming earlier research on business cycles and entrepreneurship (Rampini, 2004; Konon et al., 2018).

In summary, our paper makes three contributions. We introduce a methodological approach into entrepreneurship by explicitly identifying age, period (year), and cohort effects. A clear characterization of these effects strengthens the empirical foundation for theory and policy. Second, we present novel evidence on generational differences in entrepreneurship by quantifying how younger cohorts compare to older ones, providing stylized cohort profiles that advance both theory and empirical work on how generational aspects shape entrepreneurship. Previous debates have often relied on estimates confounded by age or cohort effects; consequently, policy recommendations based on such evidence risk being misdirected. Third, this paper employs a novel large-scale and harmonized cross-country dataset to investigate entrepreneurship, thereby complementing single-country studies (*e.g.*, Xiao and Wu, 2021; Hatak and Zhou, 2021; Jayawarna et al., 2014) and ensuring replicability of

our findings.

2. Theoretical considerations

Entrepreneurial activities – including entry, active entrepreneurship, and exit – are closely linked to an individual’s age (Kautonen et al., 2017; Lévesque and Minniti, 2006; Welter, 2011). Previous research extensively discusses the theoretical reasons and provides hypotheses on why we should expect an inverted U-shaped relationship between age and entrepreneurship, peaking at midlife (see, *inter alia*, Kautonen et al., 2014; Xiao and Wu, 2021; Wickstrøm et al., 2022). Therefore, we briefly explain key arguments building on the literature on human, social, and financial capital formation.

We will then discuss how cohort effects shape the age–entrepreneurship profile. We examine how the average propensity to engage in entrepreneurship may change across cohorts and how the slope of the age profile may vary by cohort. To demonstrate how both types of cohort effects moderate age effects, we examine how institutional changes across cohorts affect the accumulation of human, social, and financial capital. We will also touch the emergence of year effects. However, since this has been subject of advanced empirical literature, we regard this as a consistency check. We conclude with a brief discussion on the emergence of cross-country differences.

2.1. The role of human capital, social capital, and financial capital endowments for age effects

Individuals make occupational choices subject to the opportunities and constraints they perceive throughout their lives. The ability to recognize entrepreneurial opportunities, as well as the constraints on doing so, depends on the accumulation of various forms of capital over the life cycle, and is a central theme in the literature on the determinants of entrepreneurship (Alvarez and Busenitz, 2001; Mary George et al., 2016; Villanueva et al., 2012; Jayawarna et al., 2014; Hoogendoorn et al., 2022). These include general and specific human capital, physical and social capi-

tal, as well as financial capital.²

Job-specific human capital consists of aspects that make an individual productive in a specific job (Schultz, 1961; Becker, 1991; Mincer, 1993). Typically, job-specific human capital is acquired on the job and can be thought of as both formal training (e.g., courses or certifications) and informal on-the-job training (e.g., experience), which correlates with age. The greater the job-specific human capital individuals possess, the better they can identify market niches, conceive ideas to improve existing products or create new business concepts (Marvel et al., 2016), and understand relevant markets and competitors (Gielnik et al., 2012; Azoulay et al., 2020).

General human capital encompasses all skills that make individuals productive across different environments (Schultz, 1961; Becker, 1991; Mincer, 1993). One dimension of general human capital is cognitive skills which can be distinguished into two broad categories: *fluid and crystallized intelligence*. Fluid intelligence refers to aspects of cognitive reasoning that require minimal prior learning, while crystallized intelligence includes learned procedures and knowledge (Cattell, 1963). Both levels vary strongly with age. At the beginning of one’s life, on average, both crystallized and fluid intelligence are developing. While crystallized intelligence increases steadily with age — albeit at an increasingly smaller rate — fluid intelligence peaks in midlife and declines thereafter (e.g. Kanfer and Ackerman, 2004; Salthouse, 2009).

Social capital, such as social networks, is another essential resource for the success and survival of businesses (Bosma et al., 2004; Aldrich and Zimmer, 1986a; Granovetter, 1973; Davidsson and Honig, 2003a): initially it tends to increase with age, until it subsequently decreases. Social networks are crucial for mobilizing resources in terms of information, potential customers, and financial capital. Entrepreneurs with strong social ties are more likely to attract the attention of investors and secure stable financing as well (Stuart and Sorenson, 2005).

²Note that we will only briefly touch on some of the main aspects that allow deriving an inverse U-shaped relationship between age and entrepreneurship. There are more reasons (see, e.g., Andric et al., 2024; Zhao et al., 2021) in an individual’s life cycle that influence, among others, opportunity recognition, which we are not able to discuss here.

Financial capital is typically obtained through an individual’s own savings or bequests, or through external finance by accessing capital markets, such as loans or equity. Personal savings typically follow an inverse U-shape over the lifespan (Modigliani and Brumberg, 1954; Ando and Modigliani, 1963; Carroll, 1997), while bequests often occur at the beginning of the second half of one’s life (*e.g.* DeNardi et al., 2010; Banks et al., 2004; Krueger, 2003). All three of them — *i.e.*, savings, bequests, and loans — may increase available financial capital directly or lead to a transfer of business ownership across generations. An individual’s ability to qualify for funding also depends on factors that typically vary with age. Financial institutions, such as banks and VC investors, often require some form of specific human capital, such as on-the-job experience. Furthermore, a creditor’s working life must be long enough to reap the returns on the associated investment.

Figure 1: Stylized contributions to the age profile of entrepreneurship

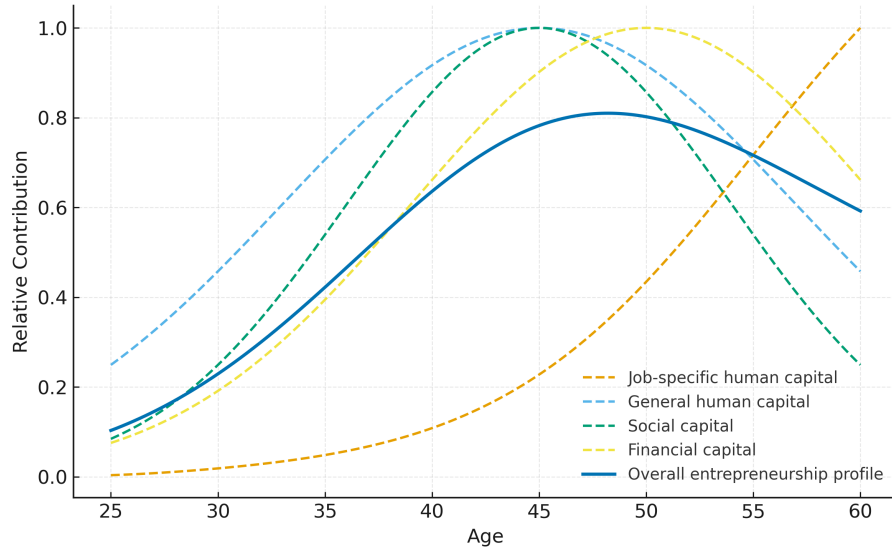


Figure 1 and Table B.1 in Appendix B summarize the mechanisms discussed above. Integrating these dynamics suggests that entrepreneurship peaks around middle age. This is when relevant resources (human and social capital, as well as financial capital) have been accumulated and can be mobilized, and when opportunities are recognized. After midlife, entrepreneurship is expected to decrease as cognitive abilities — and the applicability of accumulated work experience —

as well as access to finance, eventually decline in older age and become constraints, counterbalancing the mostly positive effects associated with higher levels of resource availability. Therefore, we test the hypothesis:

Hypothesis 1. *The age profile takes the form of an inverted U-shaped relationship between age and entrepreneurial activities, with a peak in mid-life.*

Note that we do not claim novelty for this hypothesis. Rather, the theory and hypothesis are intended to clarify how we conceptualize age effects distinct from cohort and year effects.

2.2. The role of institutions and economic transformations

As entrepreneurial decisions are based on perceived opportunities and constraints, they also depend on the specific contextual factors in which the individual lives (Welter, 2011). Contextual factors exist in many dimensions, among them with respect to differing cohorts (generations, see Zhang and Acs (2018)), differing time periods (business cycles, Clementi and Palazzo (2016); Konon et al. (2018)), and across countries (Urbano et al., 2019).

2.2.1. Differences between cohorts

Individuals born in the same year experience similar historical environments shaped by common institutions such as education systems, labor markets, capital markets, and modes of communication (Mannheim, 1952; Glen H. Elder, 1974; Ryder, 1965). These institutions affect incentives and preferences (Alesina and Fuchs-Schündeln, 2007), which, in turn, influence how individuals accumulate various forms of capital (see Section 2.1). The early years of life are often termed the “impressionable years,” a period during which attitudes are particularly malleable. During this time, exposures are expected to have particularly strong effects on individuals (Krosnick and Alwin, 1989; Sears and Levy, 2003; Alwin et al., 1991). Notably, a growing body of economic literature shows that these formative influences extend beyond

norms, values, and political attitudes to also shape economic outcomes ([Alesina and Fuchs-Schündeln, 2007](#); [Fisman and Schneider, 2008](#); [Malmendier and Nagel, 2011](#)).

We argue that cohort effects may manifest in two ways. In standard treatments, they appear as level shifts in the outcome variable, generated through shared exposure to institutions and common experiences—such as differing levels of human capital at the end of schooling, macroeconomic conditions during labor market entry, or heterogeneous initial endowments of financial capital at the beginning of one’s working life. However, these experiences are also likely to affect the trajectory of the age–entrepreneurship profile, not just its level. In the following, we describe the sources of cohort-related differences across various dimensions of capital.

Job-specific human capital. Labor markets have changed over time. For instance, in many advanced economies, job stability has declined, leading to shorter tenure spells ([Goulart and Oesch, 2024](#); [Bussolo et al., 2022](#); [Molloy et al., 2024](#)). As a consequence, incentives have shifted from accumulating highly job-specific human capital toward more transferable forms. In turn, both the baseline level and the slope of the age–entrepreneurship profile may have shifted across cohorts, as individuals increasingly accumulate skills applicable across different work environments, including entrepreneurial activity.

General human capital. Long-run data on educational attainment show that general human capital has steadily increased over the past two centuries ([Lee and Lee, 2016](#)). Moreover, the second half of the twentieth century witnessed a marked expansion in tertiary education participation across countries ([Schofer and Meyer, 2005](#)). Unlike specific human capital, general human capital—proxied by school-leaving credentials—typically plateaus at about age 25, at which point individuals have acquired most formal educational credentials. Furthermore, individuals with more schooling tend to exhibit higher cognitive skills (*e.g.*, [Carlsson et al., 2015](#)). Consequently, significant educational reforms are likely to contribute to cohort differences. In particular, education has been shown to increase the likelihood of engaging in entrepreneurial activities (*e.g.*, [Ahn and Winters, 2023](#); [Fritsch et al., 2015](#)).

Social capital. The way individuals socialize and interact has also changed over time. Changes in modes of communication can have persistent effects on the structure and scope of individuals' social networks. A prominent example is the introduction of social media platforms (Putnam, 2000; Ellison et al., 2007; Burt, 2005). This development may appear as both a level shift and a change in the age profile, as individuals are able to build and maintain broader business networks more quickly, through business related social media platforms.

Financial capital. Another important development is the shift from industrial production toward the service sector. Whereas manufacturing often requires substantial up-front investment to enter, operating in the service sector typically entails lower up-front costs (Fourastié, 1949; Baumol, 1967; Schettkat and Yocarini, 2003; Herrendorf et al., 2014; Autor, 2015). Further, financial markets underwent a strong push toward liberalization in the second half of the twentieth century. This changed the channels through which individuals can acquire financial capital (Black and Gilson, 1998; Greenwood and Scharfstein, 2013). Again, these forces may shift both the baseline probability of entrepreneurship and the slope of its age profile.

Figure 2: Cohort-specific age slope (multiplicative cohort effects)

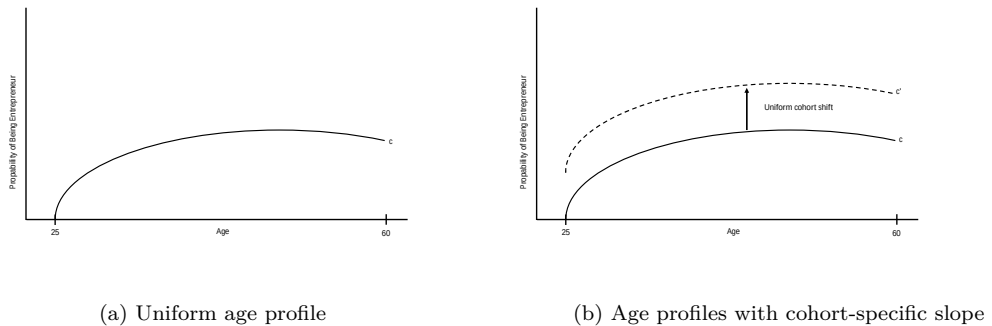


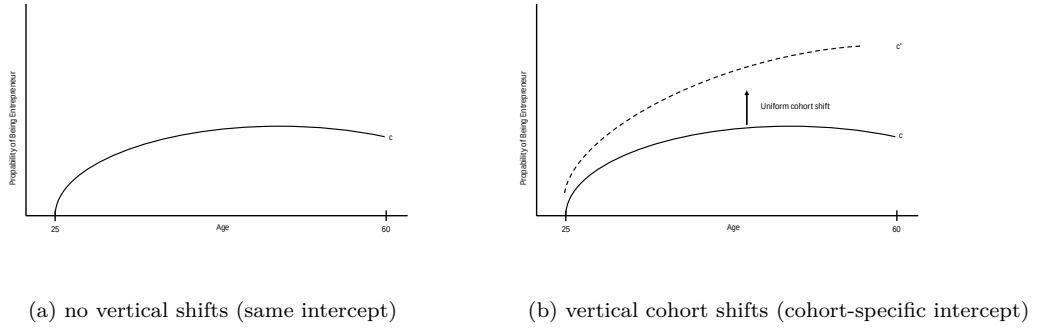
Table B.2 in Appendix B summarizes the previous arguments. Integrating these findings, we expect that cohorts will differ in their propensity to engage in entrepreneurial ventures, in both, the slope of the age-entrepreneurship relationship as well as in the average level. On this basis, we hypothesize that

Hypothesis 2. *Each cohort has its own specific age profile of entrepreneurship, i.e.,*

the slope of the age-entrepreneurship relationship differs between cohorts (Figure 2).

Hypothesis 3. *The probability of being an entrepreneur increases between cohorts, leading to vertical shifts in the age curve between cohorts (Figure 3).*

Figure 3: Vertical shifts (additive cohort effects)



To empirically assess our three hypotheses, we will disentangle age, period, and cohort effects. We then examine the extent to which an inverted U-shaped relationship exists between age and entrepreneurship (*hypothesis 1*). We will also test for interaction effects between age and cohort. If no interaction effects are present, the derived age profiles are common across cohorts, differing only by vertical shifts induced by additively separable cohort effects (*hypothesis 3*).³ If age and cohort effects interact (*hypothesis 2*), the separability assumption does not hold, though the derived age profiles can still be interpreted as average profiles across cohorts. While informative, these average profiles may obscure heterogeneity across cohorts.

2.2.2. Differences by time period

There is a robust body of literature on the relationship between business cycles and entrepreneurship—i.e., on year effects, particularly how changes in economic growth influence entrepreneurial activity. Theoretically, there are sound reasons to expect pro-cyclical effects of changes in GDP on entrepreneurship, whereby total demand

³Note the differences between *hypotheses 2* and *3*. While *hypothesis 2* posits that the slope of the age curve varies across cohorts, *hypothesis 3* suggests that cohort differences manifest in the intercepts—i.e., the entire curve shifts upward or downward depending on the cohort.

drives entry and exit. Accordingly, one would expect increased entrepreneurial activity during economic upswings and decreased activity during recessions (Rampini, 2004; Barlevy, 2007). There is also support for counter-cyclical effects: Francois and Lloyd-Ellis (2003) argue that resource costs are lower during recessions and, since entrepreneurial activity is sensitive to entry costs, it may increase during downturns. Determining which effect predominates is an empirical question. Research reports divergent findings: pro-cyclical effects in the US (Clementi and Palazzo, 2016), counter-cyclical effects in Germany (Konon et al., 2018), and more indirect counter-cyclical effects in the UK (Georgellis and Wall, 2000).

Hypothesis 4. *The probability of being an entrepreneur is subject to year effects.*

2.2.3. Differences between countries

Another strand of literature reveals significant cross-country variation in entrepreneurial activity (*e.g.*, Urbano et al., 2019). Differences are attributed to variations in formal institutional contexts across countries—such as regulatory frameworks—that can facilitate or constrain access to resources (Djankov et al., 2002) and provide the economic freedom necessary to start and operate a business (McMullen et al., 2008; Kraft et al., 2022). They are also linked to differences in informal institutions, such as values, norms, and entrepreneurial culture—each of which can either support or hinder entrepreneurship (North, 1990; Urbano et al., 2019). Social norms and societal belief systems also influence individuals’ human capital endowments and, consequently, their opportunity perception, motivation, and risk attitudes.

Furthermore, institutions differ across countries in terms of access to external financing instruments, taxation systems (insofar as they affect entrepreneurial activity), labor and product market regulations, and the structure of the entrepreneurial ecosystem, to name just a few (Ciccone and Papaioannou, 2007; Levie and Autio, 2011; Stam and Van de Ven, 2021). Building on these insights, we argue that individuals’ perceptions of entrepreneurial opportunities and constraints vary substantially depending on their exposure to country-specific institutional contexts. However,

since we limit our analysis to three countries in order to examine the generalizability of our findings, we refrain from formulating a hypothesis at the country level.

3. Empirical setting and methods

3.1. Data sets

We use rich individual-level panel data from three countries. The Comparative Panel File (CPF) is a unique open science project that harmonizes the largest and longest-running representative household surveys from various countries. It comprises detailed information about the socio-economic situation of household members, including work history, education, and attitudes. In its raw form, it includes about 2.7 million observations from roughly 360,000 individuals. For our analysis, we use CPF version 1.5 (Turek et al., 2021).

We select three countries: Germany, the United Kingdom, and the United States. The rationale for doing so is that these are prominently featured in the literature on entrepreneurship and the household surveys in these countries have been running for several decades, providing a sound empirical basis.⁴ Table 1 lists the data sources harmonized in the CPF, alongside the reference article for each country in our analysis.

Table 1: Description of Data Sets

Country	Years	Data Set	Main Reference
Germany	1984–2020	Socio-Economic Panel (SOEP)	Goebel et al. (2019)
UK	1991–2008	British Household Panel Survey (BHPS)	Buck and McFall (2012)
UK	2009–2021	UK Household Longitudinal Study (UKHLS)	Buck and McFall (2012)
US	1968–2019	Panel Study of Income Dynamics (PSID)	McGonagle et al. (2012)

⁴The other four countries, which we do not use here, are Australia, Russia, South Korea, and Switzerland. The panels in these countries begin at the earliest in the late 1990s and are, therefore, less suitable for life cycle analyses across cohorts.

3.2. Variable definitions and sample construction

Dependent variable

Entrepreneurship, our main outcome in this analysis, is operationalized as follows: For each respondent, we create an indicator set to one if the respondent reports being founder, owner, and/or manager of their own for-profit firms at their own risk to generate income and zero otherwise, using the harmonized variable `selfemp` in the CPF v1.5. As we are interested in this status irrespective of the age of the firm, entrepreneurs could be running their firm for months or years. Concentrating on the status (and not restricting the analysis to the entry into entrepreneurship) is the necessary condition to be able to analyze our research question, as we aim to capture entrepreneurial activity, which we defined as “entering into, maintaining, and exiting from entrepreneurship and self-employment,”. In that sense, we take a broad view how we describe entrepreneurship, (see, *e.g.*, also [Lévesque and Minniti \(2006\)](#) and [Zhao et al. \(2021\)](#)). To capture the full set of outside options, the alternative consists of paid employment or no employment. Therefore, not being self-employed includes employees and individuals who are either unemployed or inactive, *i.e.*, the complete workforce potential, as individuals who are not employed, could become entrepreneurs at any time.

Independent variable

We measure *age* as the difference between the year and an individual’s year of birth. In our main analysis, we focus on people aged 25 to 60 years to avoid spurious findings emerging through individuals entering the labor market from education or individuals leaving the labor market via retirement (see, *e.g.*, also [Fitzenberger et al. \(2021\)](#) for using this age range). Specifically, we expect the data-generating process around and in the retirement age to differ from that during the prime labor market age. However, we include a sensitivity analysis where we extend the age range up to 75 years, thus also capturing late-career entrepreneurship ([Kautonen et al., 2017](#)).

Data cleaning

We exclude all observations with missing information on employment status. Furthermore, we exclude a new sample of top-income earners in Germany added to the SOEP in 2012 and 2019 because they have a disproportionately high share of entrepreneurs (12% versus an average of 5%), and two additional samples of refugees in 2016 and 2017, since refugees typically do not have permission to work for extended periods. For the US, new samples added in 1990 (Hispanics) and 1997 (immigrants) show no diverging pattern with respect to entrepreneurial activities compared to the other samples, so these are included. The same holds true for the new UK samples on ethnic minorities and immigrants added in 2009 and 2015. As a result, we include these additional samples for the UK estimations.

To ensure a common time span, we restrict our sample start to 1992 after German reunification.⁵ Since 1998, the PSID surveys US households biennially, with the most recent PSID year available in the CPF v1.5 being 2019. Therefore, our final sample includes observations ranging from 1992 until 2019. Table 2 lists summary statistics.

Table 2: Number of Observations

Country	Time Span	Number of Individuals	Number of Observations	Avg. Self-Employment Rate (1992–2019)
Germany	1992–2019	55,628	372,958	7.1%
UK	1992–2019	67,147	408,901	10.6%
US	1992–2019	29,149	174,015	9.5%
Sum		151,924	955,874	

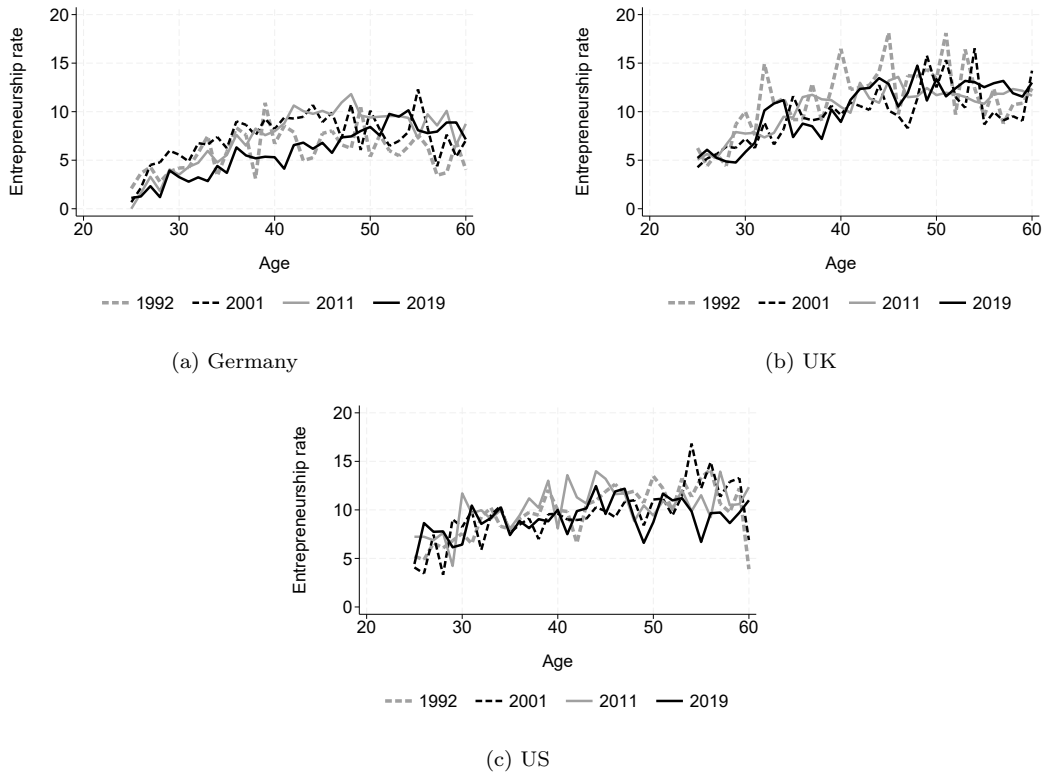
Source: Own calculations based on the CPF v1.5.

⁵For the period before German reunification, the SOEP only contains data for the West German population. Furthermore, the UKHLS has only very few observations on UK households in 1991.

3.3. Descriptive results

Figure 4 displays the age curves in the raw data for selected cross-sections (1992, 2001, 2011, and 2019). The age curve of being an entrepreneur varies greatly between the different cross-sections. Furthermore, there are cross-country variations in terms of the initial share of entrepreneurial activity at young ages (*i.e.*, at 25 years) and in the course of the age–entrepreneurship relationship over the life cycle.

Figure 4: Entrepreneurship Rates by Age in Selected Years



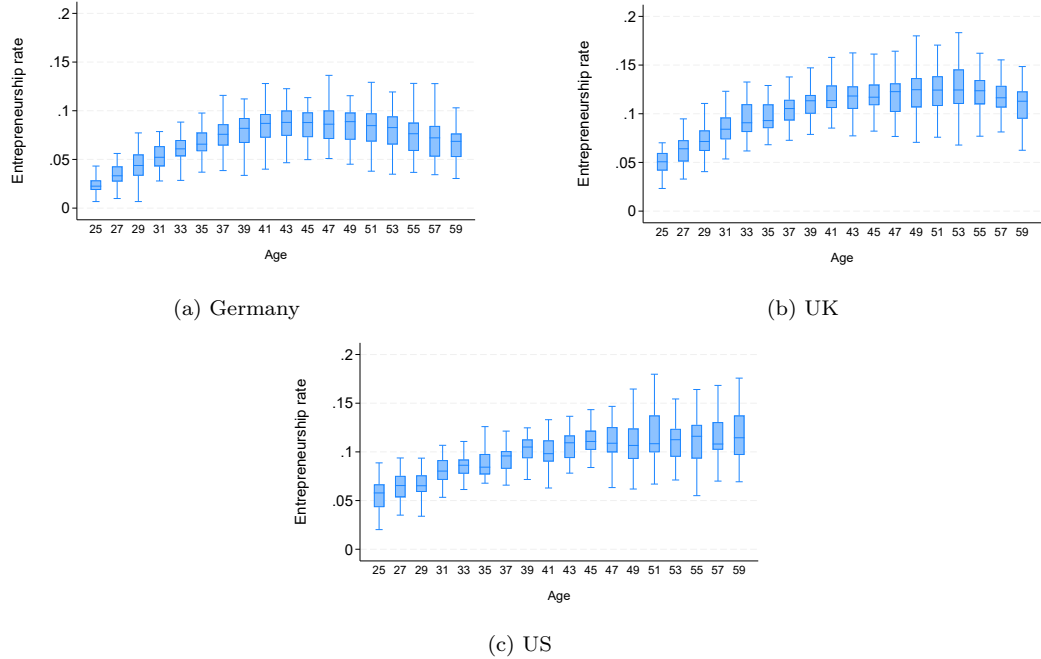
Note: Figures 4a to 4c illustrate the variation of the age profile at different points in time (cross-sections) in the sample. Own calculations based on CPF v1.5.

However, without any identifying assumptions, a simple cross-sectional depiction of entrepreneurship rates by age group is insufficient to infer the actual age profile of entrepreneurship. For any given calendar year, observed differences in entrepreneurship rates across age groups reflect both age *and* cohort effects. The reason is that, for any given cross-section, as one moves from the average probability of being an

entrepreneur at one age to the next, individuals are not only one year older but also come from a birth cohort that is one year earlier. For that reason, it is impossible without any further assumptions to infer the age profile. One implicit assumption in previous empirical studies so far, which is rarely explicitly stated, is that there exist no cohort differences in the average probability of being an entrepreneur. As we will show, this assumption is not supported by our results.

Likewise, within a given cohort, observed variation in entrepreneurship rates by age conflates age and year effects. As members of a cohort grow one year older, they are also observed one year later in calendar time, so any change in their entrepreneurial activity may reflect aging, contemporaneous shocks, or both. This mechanical confounding complicates the interpretation of within-cohort age patterns unless one adopts explicit identifying assumptions or a design that separates age from calendar year effects.

Figure 5: Entrepreneurship Rates by Age (Boxplots)



Note: Figures 5a to 5c illustrate the variation of the age profile across all cohorts in the sample. For example, the boxplot at age 25 shows summary statistics (25-quantile, 75-quantile, and median) for the entrepreneurship rate at age 25 between 1992 and 2019, i.e. across all cohorts born between 1967 and 1994. Age grouped in two-year intervals. Graphs exclude outside values. Own calculations based on the CPF v1.5.

Figure 5 illustrates this point empirically by plotting age profiles for every cohort in our sample: the figure reveals substantial dispersion across cohort-specific age curves, underscoring that average age patterns hide important heterogeneity. One salient descriptive finding is that entrepreneurial prevalence at young ages differs markedly across countries: at age 25, the US and the UK exhibit average entrepreneur shares of roughly 5% of the population, whereas Germany averages only about 2.2% for 25-year-olds over 1992–2019 (Figure 5). These contrasts demonstrate why simple cross-sectional or pooled summaries may be misleading and motivate our identification strategy.

3.4. Empirical Strategy

Our analysis aims to disentangle age, cohort, and year effects across the three countries. The primary challenge in identifying age profiles arises from the perfect linear dependency among age a , calendar year t , and cohort c , given by the identity $t = c + a$. In our setting, the cohort corresponds to an individual’s birth year. Consequently, an individual’s age a is simply the difference between the calendar year t and the birth year c . Because age can be directly inferred from birth and calendar year, it becomes inherently impossible to separately identify the effects of age, calendar year, and cohort without additional identification assumptions (*e.g.*, Mason et al., 1973; Glenn, 1976; Heckman and Robb, 1985; Yang and Land, 2006).

Section 2.2 provides theoretical insights for interpreting these effects. For example, *year effects* could capture fluctuations in entrepreneurship rates stemming from macroeconomic conditions, such as business cycle variations or major economic disruptions like the 2008–2009 financial crisis. *Cohort effects* reflect structural shifts influencing specific birth cohorts. Examples discussed in Section 2.2.1 include societal transformations such as the post-World War II expansion of educational opportunities in various OECD countries. *Age effects* refer to factors associated with age, such as job-specific human capital or cognitive aging.

To address this identification challenge, we impose parametric assumptions, fol-

lowing the methodologies of [Fitzenberger et al. \(2021\)](#) and [MaCurdy and Mroz \(1995\)](#). Specifically, we estimate the following empirical model to identify the age profile of entrepreneurship in each country:

$$y_{ict} = G_i + \gamma_{2i}c_{it}^2 + \cdots + \gamma_{pi}c_{it}^p + \alpha_{1i}a_{ict} + \alpha_{2i}a_{ict}^2 + \alpha_{3i}a_{ict}^3 + D^t\beta_i + u_{ict}. \quad (1)$$

In Equation 1, y_{ict} represents the probability of being an entrepreneur for cohort c in country i in year t . Following [Fitzenberger et al. \(2021\)](#), we compute mean entrepreneurship rates for each cohort-year cell, and use the number of observations in each cell as regression weights. Aggregating individual-level data into synthetic cohorts addresses the challenge that the available panels are typically too short to track individuals across their full life course. Furthermore, it mitigates bias from non-random attrition.

We estimate Model 1 separately for each country, such that the final level of observation within each country i is cohort c in calendar year t , with the number of observations $N_i = C_i \times T_i$. The term $G_i = a_0 + c_0 + t_0$ is a constant that can be interpreted as the probability of being an entrepreneur for the baseline cohort at the baseline age in the baseline calendar year. As we consider all individuals between the age of 25 and 60 observed between 1992 and 2019, the baseline age is $a_0 = 25$ and the baseline cohort c_0 comprises all persons born in the year 1967 ($c_0 = t_0 - a_0 = 1992 - 25 = 1967$). Note that, as a result of this identification, we discuss age, year, or cohort effects relative to the respective baseline category. Therefore, the estimated age, cohort, or year effect does not inform us about the level of the probability of being an entrepreneur, but rather the difference relative to the baseline category.⁶ Throughout, we cluster standard errors at the cohort level to account for within-cohort correlation. To improve the reliability and reproducibility

⁶Note that this is not a limitation due to our identification assumption, but inherent to all regression models in which the intercept typically corresponds to some baseline category.

of our findings, we adopt the significance thresholds suggested by Benjamin et al. (2018). Specifically, we refer to findings with $p < 0.005$ as significant and $p < 0.05$ as suggestive evidence, where p denotes the probability of obtaining test results at least as extreme as those actually observed, assuming the null hypothesis is correct.

Age Effects. Model 1 includes a third-order polynomial in age a .⁷ Based on the estimates for coefficients α_1 , α_2 , and α_3 , we calculate predictions that reflect the probability of being an entrepreneur at a certain age relative to the baseline age a_0 . These predictions constitute the age effects, which add up to the age profile. To characterize this age profile statistically — including testing for its curvature — we follow Lind and Mehlum (2010) and Haans et al. (2016) by analyzing three aspects: (i) the estimated coefficients for the age polynomial; (ii) the slope at the curve’s boundaries; (iii) the turning points.

Year Effects. To capture year effects in a flexible and non-parametric way, we include year dummies D^t to estimate calendar year effects, omitting the first year of observation—*i.e.*, the reference year 1992.

Cohort Effects. Intuitively, a uniform cohort shift increases the probability of being an entrepreneur uniformly for all individuals of the same birth cohort. The age profile is shifted vertically, without affecting its slope (see Section 2.2.1, Figure 3). We estimate the cohort-specific shifts in the entrepreneurship rate using a 5th-order polynomial in the year of birth, denoted c_{it}^p in Model 1 with $p = 5$. The predictions based on the estimates for γ_2 to γ_5 are the cohort effects. They correspond to the cohort-specific shifts in the probability of being an entrepreneur, relative to the baseline cohort c_0 .

Identification. As mentioned at the beginning of this section, the age–period–cohort (APC) relation is perfectly collinear because cohort = period (year) – age. To achieve separate estimation of these three components, we omit the linear-in-cohort

⁷We chose a third-order polynomial to allow for a flexible parametric approximation of the true underlying data-generating process. However, we also display non-parametric estimates, based on a full set of estimates associated with age indicators. Throughout, results are consistent across specifications.

term c_{it} ; omitting this linear trend breaks the exact linear dependence. This step is critical: without it, observed patterns cannot be unambiguously attributed to aging, cohort-specific experience, or contemporaneous shocks, and empirical descriptions of age profiles risk being fundamentally misleading. Making this identifying choice explicit thus provides a stable empirical foundation for theoretical interpretation and policy recommendations.

4. Results

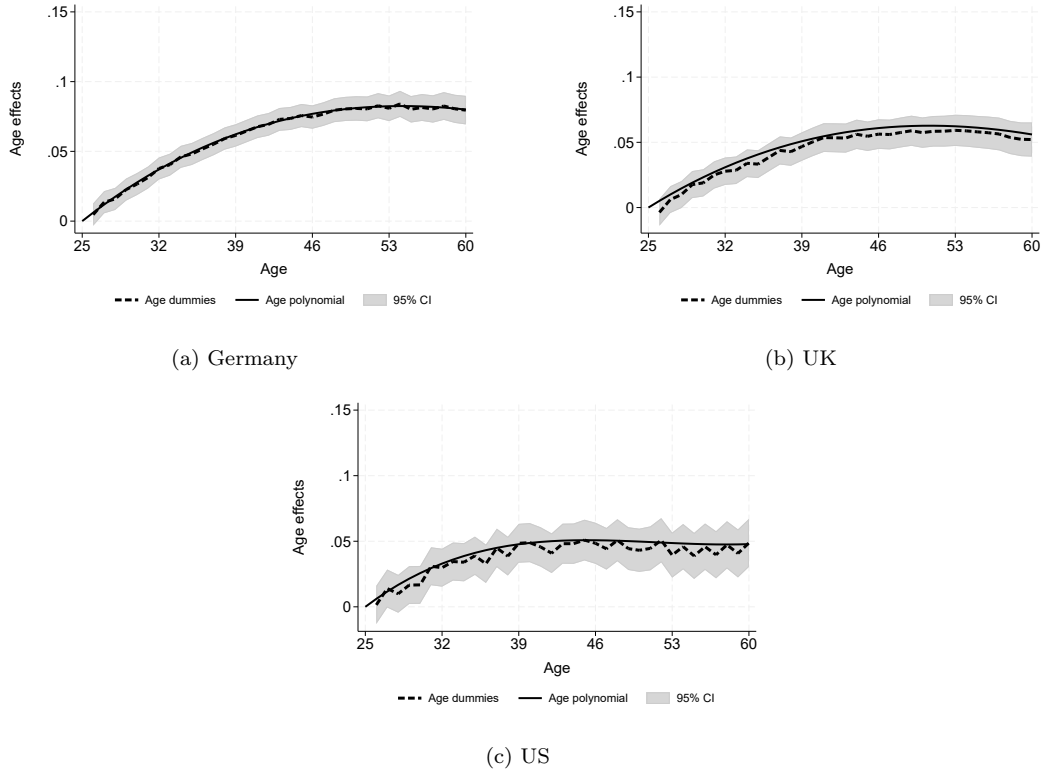
4.1. Age Effects

4.1.1. 25 to 60 years

Figure 6 illustrates the estimated age profiles for Germany, the UK, and the US. In all three countries, the probability of engaging in entrepreneurial activity increases with age until reaching a plateau during middle age. Table 3 provides additional details on the slope of the age profile and identifies the respective turning points. Specifically, column (1) tests whether the slope of the age profile differs significantly from zero on the left-hand side of the turning point, while column (2) does so for the right-hand side. Column (3) reports the estimated turning point—the age at which the probability of entrepreneurial activity peaks. Detailed estimation results for the age profile in each country are also reported in Table B.3 in Appendix B.

The results reveal notable cross-country differences in both the magnitude of the slope and the turning points. In Germany, we observe the steepest curve: relative to the baseline (age 25), the probability of being an entrepreneur increases by approximately 8 percentage points up to the turning point at age 48 (Table 3, column (3)), after which it remains relatively stable. By contrast, in the UK, the probability increases by about 6 percentage points until the turning point at age 46.4, remaining at an approximate level of 6% thereafter. In the US, the age profile rises at a rate similar to that of the UK but peaks earlier at age 45.2, with the probability stabilizing at around 5 percentage points, relative to the baseline age group.

Figure 6: Common age profile



Note: Figures 6a to 6c illustrate the age profile of the probability of engaging in entrepreneurship for Germany, the UK, and the US. The black line shows predicted age effects after estimating Model 1 with a third-order polynomial in age. The dashed lines correspond to the predictions from estimating Model 1 using a full set of age indicators. The shaded areas correspond to 95% confidence intervals based on standard errors clustered by cohort. Own calculations based on the CPF v1.5.

Overall, the findings highlight significant variation in the probability of engaging in entrepreneurship across age, both within and across countries. Consequently, we partially confirm Hypothesis 1: the entrepreneurship rate steadily increases until middle age—peaking somewhat before age 50, depending on the country. However, contrary to expectations, it does not decline immediately after age 50 but remains on a plateau for at least 10 years.

Table 3: Tests for the shape of the age profile

	steepness at the tails		turning points	
	left tail	right tail	a_{tp} s.t.	
	$H_0 : f'(a_L) = 0$	$H_0 : f'(a_R) = 0$	$f'(a_{tp}) = 0$	
Germany	0.000***	0.035*	48.0	
UK	0.000***	0.077	46.4	
US	0.000***	0.698	45.2	58.0

Notes: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Table 3 reports the results for the inverted U-shape tests following Lind and Mehlum (2010). The first column reports the p-value from testing the null hypothesis that the first derivative of the age polynomial is zero at age 26. The second column reports the p-value from testing the null hypothesis that the first derivative of the age polynomial is zero at age 60. The third and fourth columns report the turning points of the estimated age polynomial, *i.e.*, the age level for which the first derivative of the age polynomial is equal to zero.

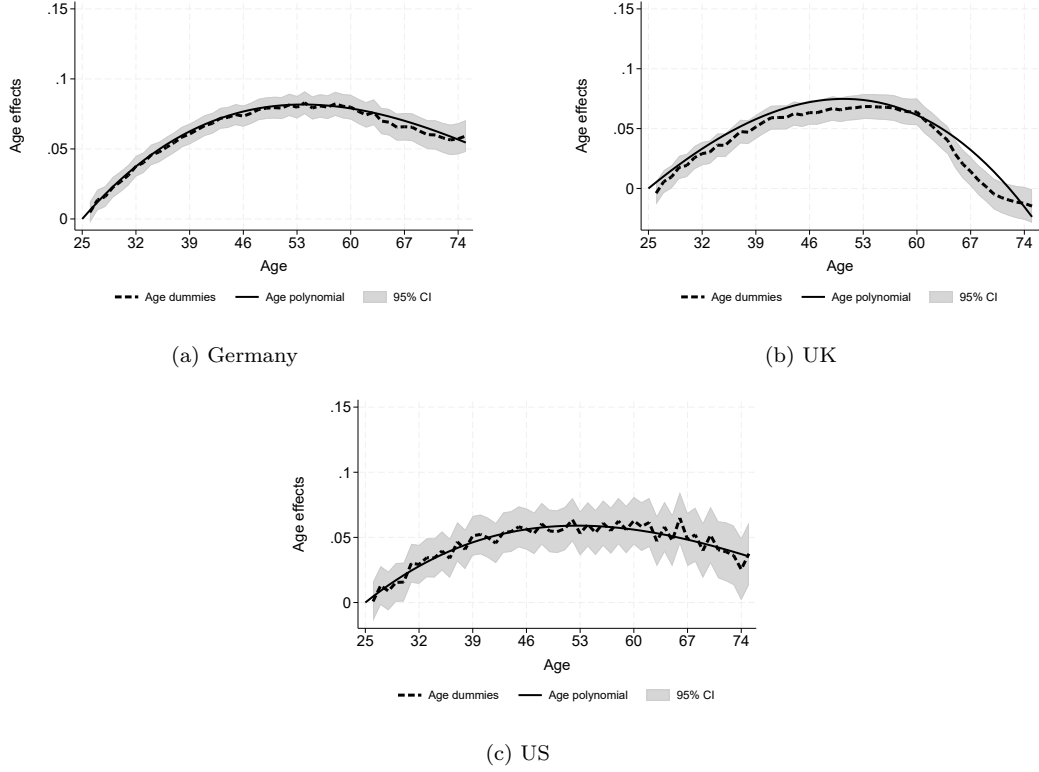
4.1.2. 25 to 75 years

Our primary analysis focuses on individuals aged 25 to 60 to avoid potential artifacts arising from transitions out of education or into retirement. However, given that many individuals continue entrepreneurial activities beyond the statutory retirement age (Kautonen et al., 2017), we expand the sample in the subsequent analysis to include individuals up to 75 years old.

In this extended analysis, two key observations emerge. First, the age profiles across cohorts reveal that the traditionally documented inverted-U shape becomes more pronounced when the population between 60 and 75 is included (Figure 7). Formal tests on the gradients indicate that the age curve significantly declines from age 70 in both Germany and the UK, whereas in the US, a significant downward slope is observed only from age 75 (see Appendix B, Table B.5, second row).

Second our results indicate that, in Germany and the US, the proportion of individuals engaged in entrepreneurship at age 75 is higher than among individuals aged 25. In contrast, the UK exhibits a pronounced decline after age 60, with the probability of being an entrepreneur 2.4 percentage points lower for 75-year-olds than for 25-year-olds. Consequently, the UK displays the most pronounced inverted-U-shaped age profile.

Figure 7: Average age profile up to 75 years



Note: Figures 7a to 7c display the age profile of the probability of engaging in entrepreneurship for Germany, the UK, and the US when extending the age range to 75 years. The black line represents the predicted age effects estimated from Model 1 using a third-order polynomial in age, while the dashed line corresponds to predictions derived from a full set of age indicators. Shaded areas denote 95% confidence intervals based on standard errors clustered at the cohort level. Note that the separability condition is not satisfied for the age range 25 to 75, implying that each cohort follows its own age profile; the curves thus represent the average age profiles across cohorts. Computations based on CPF v1.5.

4.2. Cohort effects

In this section, we investigate cohort differences in the probability of being an entrepreneur. For this, we proceed in two steps.

4.2.1. Multiplicative cohort effects

First, we investigate whether the age profiles vary across cohorts, *i.e.*, whether there exist interactions between age and cohort effects. If this is the case, the age profiles in Section 4.1.1 correspond to average age profiles across cohorts. If no interactions between age and cohort effects exist, the estimated age profiles are

uniform across cohorts, up to cohort-specific level shifts. We test interaction effects between age and cohorts by setting up a more general model and testing for the joint significance of the interaction terms (see Appendix A for details). We run the model once for the age range of 25 to 60 years and once for 25 to 75 years. The first column in Table A.1 in Appendix A shows that the null hypothesis, *i.e.*, the absence of joint significance, cannot be rejected for the age range of 25 to 60 years, suggesting that the separability assumption is correct. In other words, all cohorts share the same slope and there exists a common age profile across cohorts in Germany, the UK, and the US between the age of 25 and 60, up to potential vertical cohort shifts.

When we extend the age range to 75....

, the separability condition—supported in our preferred age window (25–60) in Section 4.1.1—does not hold when extending the age range, indicating significant interaction effects between age and cohort. This result implies that while cohorts exhibit similar age profiles up to age 60, the rate and extent of exit from entrepreneurship after age 60 vary by cohort, formally expressed as differences in slopes (see Appendix B, Table B.4, first row).

< Final sentence wrt to Hypothesis 2: Thus, we reject Hypothesis 2 for the age range of 25 to 60 but confirm for 25 to 75?>

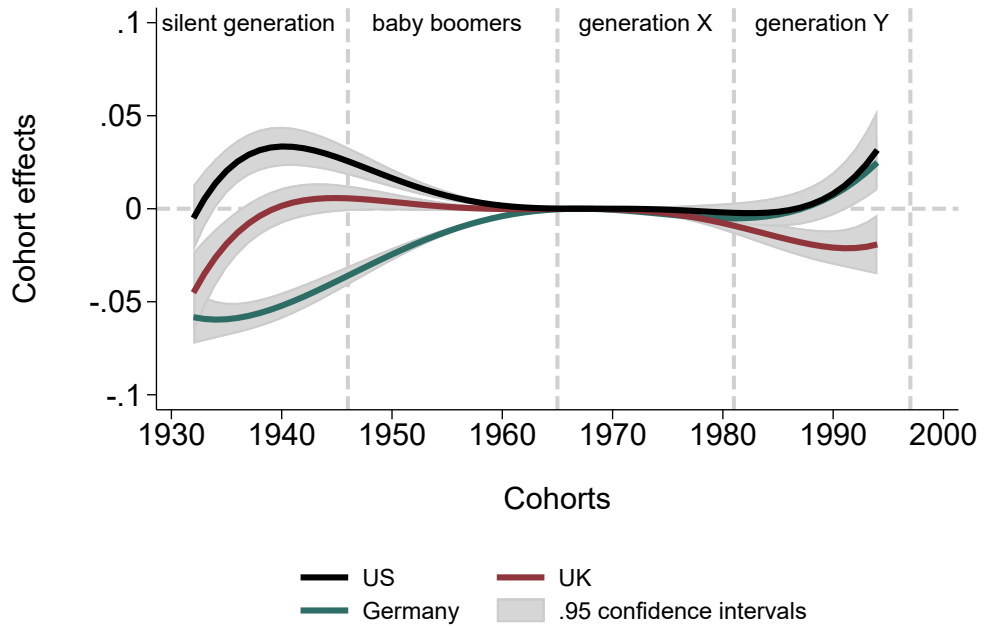
4.2.2. Additive cohort effects

Next, we examine the cohort shifts by testing whether the baseline levels of the age profiles are identical across cohorts; *i.e.*, whether the curves of the age profiles always start at the same level. We find strong evidence for additive cohort effects, that is, although they share the same slope, cohorts differ in their baseline level of entrepreneurial activity (Table A.1, column 3). Hence, Hypothesis 3 is confirmed.⁸

Figure 8 illustrates these cohort effects in more detail, with individuals born in 1967 serving as the reference cohort. The figure shows that there exists large

⁸Intuitively speaking, this could mean that persons born in 1950 were more often active as an entrepreneur at the age of 25 than those born in 1990 (*cohort effect*) but both cohorts increase their entrepreneurial activities by 5 percentage points between age 25 and 30 (*age effect*).

Figure 8: Cohort effects in the propensity to become self-employed



Source: Own calculations based on based on CPF v1.5.

variation across cohorts, especially for the cohorts born before 1967. To describe the subsequent variation efficiently, we will rely on a common classification of ranges of birth cohorts into 'generations'. However, we refrain from characterizing these since the concept of generations, as the aggregate of many birth cohorts, has been questioned (*e.g.*, [Rudolph et al., 2021](#)).

In Germany, cohort differences in entrepreneurship are most pronounced. The Silent Generation (1928–45) had the lowest probability of being entrepreneurs, where the earliest cohort in our analysis, born in 1932, was 5.8 percentage points less likely to be entrepreneurs than the 1967 reference cohort from Generation X (1965–79). The probability of being an entrepreneur steadily increased among the Baby Boomers (1946–64), before stabilizing in Generation X. This pattern of stabilization, *i.e.*, little variation in the probability of being an entrepreneur among cohorts belonging to Generation X, can be observed in all three countries. Among the Generation Y (also called Millennials) born after 1980, entrepreneurship rates began to increase again in Germany.

In contrast to Germany, cohort variation in the United Kingdom is relatively modest. Individuals from the Silent Generation born in 1932 were about 4 percentage points less likely to be entrepreneurs than those born in 1967. However, convergence occurred quickly, as the cohort born in 1936 shows no significant difference compared to the 1967 cohort. Subsequent cohorts display no statistically significant variation in entrepreneurship until a downward trend begins among Generation Y born in 1980. Notably, by 1993, the entrepreneurship rate of the cohort is 2.5 percentage points lower relative to the 1967 cohort. This significantly differs from the developments in the other two countries.

The United States present a unique pattern too. Unlike in Germany and the UK, the Silent Generation and the early Baby Boomers demonstrate a notably higher level of entrepreneurial engagement than Generation X, with the peak observed in the 1940 cohort — where individuals are approximately 3 percentage points more likely to be entrepreneurs compared to those born in 1967. After this peak, the differences converge to zero, with estimates becoming statistically indistinguishable from zero around the 1960 cohort. Thereafter, the level of entrepreneurship remains relatively stable until a renewed increase is observed among Generation Y born after 1990, similar to Germany.

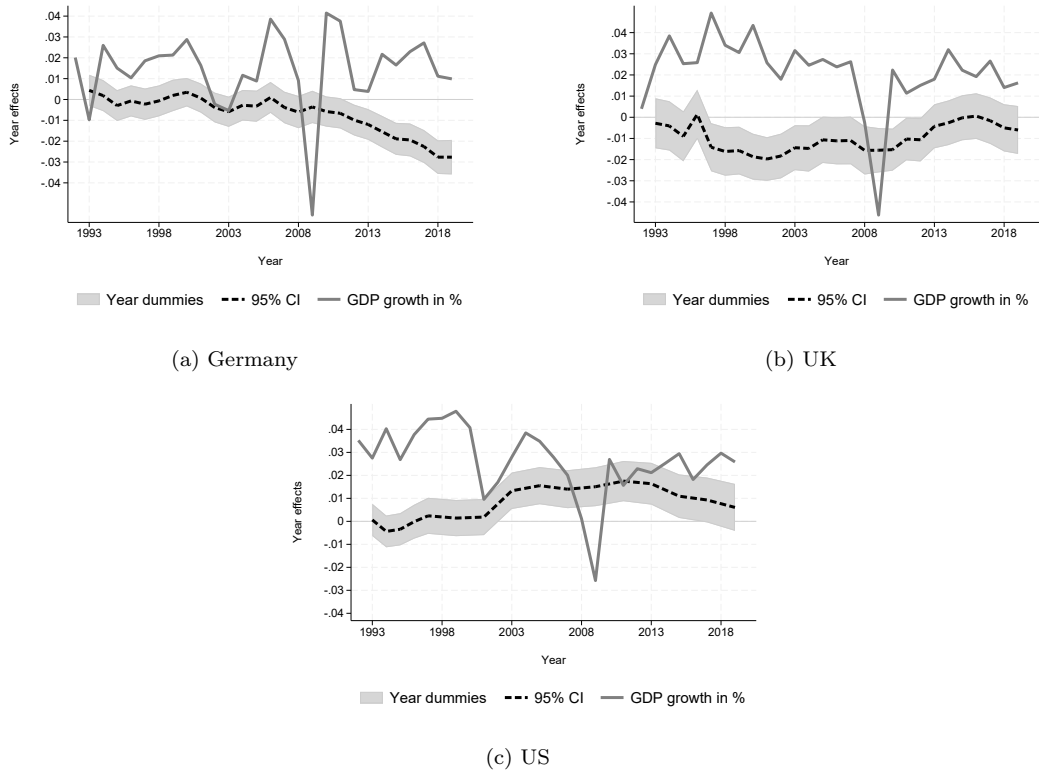
Thus, while the slopes of the age profiles in the three countries do not differ between cohorts, there are significant differences in the *level* of entrepreneurship between cohorts, *i.e.*, *cohort effects*, and these differ across the three countries. The age curves share the same slope but not the same intercept. While we are able to identify common dynamics throughout the age range between 25 and 60 years, in each of the three countries each cohort starts at an individual baseline level.

4.3. Year effects

Similar to the findings for age and cohort effects, we observe substantial variation in year effects both within and between countries. Figures 9a to 9c display the estimated year effects for Germany, the UK, and the US, along with the correspond-

ing annual country-specific GDP growth rates. In Germany, year effects remain statistically insignificant until the financial crisis of 2009, after which they turn negative and grow in magnitude. Specifically, by 2019, individuals were approximately 3 percentage points less likely to be entrepreneurs compared to the reference year 1992.

Figure 9: Year effects vs. business cycles



Note: Figures 9a to 9c compare the estimated year effects with annual GDP growth. GDP data is taken from World Bank (2025): World Development Indicators (NY.GDP.MKTP.KD.ZG). Own calculation based on the CPF v1.5.

In contrast, the UK exhibits a U-shaped pattern in year effects. Here, the probability of being an entrepreneur declines steadily, reaching its lowest point in 2001, when individuals were around 2 percentage points less likely to be entrepreneurs relative to 1992. Subsequently, the year effects rebound, increasing steadily until they become statistically indistinguishable from zero from 2013 onward.

For the US, the pattern is characterized by an inverted U-shape over most of the observational period. From 1993 to 2001, the year effects are statistically indistin-

guishable from zero; however, following 2001, they increase, reaching a peak in 2011. In that year, the estimates indicate that individuals were about 2 percentage points more likely to be entrepreneurs compared to those in the reference year 1992.

In summary, the empirical evidence is in line with Hypothesis 4 and suggests that temporal dynamics in entrepreneurship are highly sensitive to macroeconomic events and exhibit distinct patterns across countries. The pronounced decline in Germany following the financial crisis, the U-shaped trend in the UK, and the inverted U-shape in the US highlight the importance of context in shaping year effects. These further confirm earlier research that mostly ended around the financial crisis, and according to which business cycles have differing effects on entrepreneurship when comparing the three countries (Clementi and Palazzo, 2016; Konon et al., 2018).

5. Discussion, limitations and future research

The entrepreneurship literature emphasizes that the probability of being an entrepreneur changes over an individual’s life cycle. Yet, it remains unclear how the age-entrepreneurship relationship is exactly shaped and how this relationship varies across cohorts and between countries.

Our analysis identifies five key findings that shed light on these aspects. First, while the likelihood of being an entrepreneur increases with age, it reaches a plateau at middle age and remains stable until about age 60; only subsequently does it start to decrease. A significant downward trend begins around the age of 70 or 75. This further clarifies that the fully inverse U-shaped age-entrepreneurship relationship, often observed in cross-sectional analyses, only emerges beyond retirement age.

Second, interestingly, we find support for the separability assumption for the age range 25 to 60. That is, a common age profile across cohorts exists within each of the three countries between the ages of 25 and 60. This finding holds despite theoretical expectations that changes in the institutional environment (labor markets, financial markets, and social environments) may alter the life course of entrepreneurship. However, when we expand our age range beyond 60, the slope of the age profile

differs across cohorts; *e.g.*, cohorts differ in the pace and magnitude at which they exit entrepreneurship. This may be caused by changes in longevity across cohorts – specifically, if entrepreneurship, including self-employment, is a way of extending working life.

Third, comparing age profiles across countries, we find that the shape of the curves differs significantly in terms of steepness and peak age. The age curve is steepest in Germany and the UK, whereas the US exhibits a relatively flat curve. This means that age is more relevant in Germany and the UK, and least relevant in the US. These stylized facts could be rationalized by prior research suggesting that work experience strongly influences the transition into self-employment in both the UK and Germany (Evans and Leighton, 1989; Blanchflower and Meyer, 1994; McManus, 2001; Lazear, 2005). Furthermore, differences in financial systems mean that US entrepreneurs have earlier and broader access to capital markets and credit (Allen and Gale, 2000; Black and Gilson, 1998; Jappelli and Pagano, 2002; Blanchflower et al., 2003). Finally, social networks provide information, legitimacy, and financing, with weak ties being especially important in the US and strong ties in Germany (Aldrich and Zimmer, 1986b; Greve and Salaff, 2003; Davidsson and Honig, 2003b; Light and Gold, 2000). The UK, as an Anglo-Saxon country close to continental Europe, may be interpreted as an intermediate case. Finally, we observe that entrepreneurial activity plays a significant role in Germany and the US beyond the age of 70 — but to a lesser degree in the UK.

Fourth, while the slope of the age profiles does not differ between cohorts until age 60, there are significant differences in the overall *level* of entrepreneurship between cohorts (*cohort effects*). This means that the entire curve (with the same shape) is shifted upwards or downwards depending on the cohort. Overall, this finding suggests that the cohort to which individuals belong influences their likelihood of being an entrepreneur. Cohort-specific factors, such as educational institutions, social norms, and cultural influences, may shape individuals' perceptions of opportunities and constraints related to entrepreneurial activities. For instance, in Germany, there

is a consistent upward trend over the four generations, with earlier cohorts being less entrepreneurial than later ones. In the US, the high rate of entrepreneurship among the Silent Generation is particularly notable, as is the decline among the Baby Boomers. In the UK, we find relatively stable baseline levels of entrepreneurship across cohorts, but a notable drop among Generation Y. The differences between the US and Germany might be due to the economic environment. In the US, postwar cohorts entered more flexible labor markets and benefited from policies supporting entrepreneurial activity (Blanchflower, 2000; Audretsch and Thurik, 2000), whereas in Germany, vocational structures and bank-based finance limited entry into self-employment for younger cohorts (McManus, 2001). Again, the UK may be seen as an Anglo-Saxon country embedded in a European context.

Fifth, calendar year effects are generally less influential than age or cohort effects, though still economically meaningful. It is notable that the three countries rarely experienced common trends between 1993 and 2008, thereby confirming earlier research on these countries who showed that there are strongly differing effects for the years prior to the financial crisis (Clementi and Palazzo, 2016; Konon et al., 2018). These findings underscore the necessity for future research to further dissect the interplay between macroeconomic shocks and the evolution of entrepreneurial behavior in the years following the financial crisis, such as during the Covid-19 pandemic, thereby enhancing our understanding of the underlying temporal mechanisms.

With our research and these five findings, we advance the entrepreneurship literature on two fronts. The literature on the nexus between age and entrepreneurship encompasses a broad spectrum of theoretical and empirical investigations (*e.g.*, Lévesque and Minniti, 2006, 2011; Simoes et al., 2016). The prevailing intuition regarding the shape of the age-entrepreneurship relationship is grounded either in theoretical frameworks (*e.g.*, Lévesque and Minniti, 2006, 2011) or in indirect evidence provided by the inclusion of age as a control variable in empirical studies (*e.g.*, Simoes et al., 2016). In a recent meta-analysis, Zhao et al. (2021) document 1,059 articles investigating, *inter alia*, the age-entrepreneurship relationship. Despite sub-

stantial scholarly interest (Zhao et al., 2021), prior work has not disentangled age, period, and cohort effects, leaving estimated age profiles subject to bias. Thus, it is imperative to adopt appropriate identification assumptions; without them, descriptions of the age profile risk being ambiguous. Therefore, our analysis of age effects is a significant step toward more consistent insights into the age profile of entrepreneurship.

Moreover, we investigate cohort differences in entrepreneurial activities. To the best of our knowledge, no existing studies employ a consistent strategy for identifying cohort effects in entrepreneurship. We provide a detailed overview of cohort differences over time. This will serve as a foundation for future theorizing and initiate research on the circumstances that shaped generational differences. In that sense, our analysis – together with the year and age effects—offers a better understanding of why entrepreneurial activities are in decline. Existing discussions are often based on cross-sectional analyses, leading to conjectures that recent birth cohorts are less hard-working or less diligent. For Germany and for the later years also for the US, our results are clear: the decline in entrepreneurial intent cannot be explained by cohort differences but by year effects. This points toward macro-level changes – economic, political, or societal – that affect all individuals within a given time period.

Understanding these differences has important implications not only for policy-making but also for the broader entrepreneurship ecosystem. To illustrate the practical relevance, consider a typical proposal in Germany. It consists of integrating economics into school curricula to instill entrepreneurial intent, *i.e.*, increasing specific human capital. However, such a policy would be misguided, as the decline since 2010 is driven by year effects. Consequently, other policy responses are required. For instance, regulatory reforms, changes in the quality of administrative processes surrounding business formation, or improved access to financial capital could help reverse this trend. Similarly, our results carry implications for investors. Since the decline is not driven by cohort differences but by year effects, investors should

advocate for institutional improvements rather than withdrawing from the market.

Our study is not without limitations; these offer fruitful starting points for future research. First, our analyses capture entrepreneurial activity by pooling entry into, remaining in, and exit from entrepreneurship. While we argue that this approach is necessary to analyze the full scope of entrepreneurial behavior, including late-career entrepreneurship, it may obscure more fine-grained dynamics among different types of entrepreneurship, *e.g.*, high-tech entrepreneurship, solo entrepreneurs vs. growth-oriented entrepreneurs, social entrepreneurship (Hoogendoorn et al., 2022), or hybrid entrepreneurship. Future research may build on these insights to compare how age-dependent effects vary across types of entrepreneurship and countries.

Our findings offer fertile ground for future research on entrepreneurial activity. By revealing different patterns in the inverse U-shaped age curve across countries, we highlight the need for future work to examine how individuals make entrepreneurial choices based not only on their age, but also on country-specific factors. These may reflect variation in (human) capital needed to identify and exploit opportunities, as well as differences in institutional constraints. For example, our finding that the curve peaks at different ages across countries presents a promising starting point for more detailed analysis. Future research should further disentangle these effects to uncover the drivers of entrepreneurship across countries and age groups, and to better understand how entrepreneurship contributes to economic development across different institutional contexts.

6. Conclusions

In this study, we investigate how entrepreneurial activity varies across an individual’s life cycle by comparing three countries – Germany, the UK, and the US. We employ a novel age-period-cohort model to separate age effects from cohort and year effects. For our empirical analysis, we use national household surveys, harmonized by the Comparative Panel File (CPF). Our results show that entrepreneurship displays the steepest age trajectory in Germany, followed by the UK. For individuals in their

prime working years, the probability of being an entrepreneur plateaus around the mid-forties through the late fifties in all three countries, contrary to the expected inverse U-shape. Only when we extend the sample beyond retirement age does the inverse U-shape emerge. Our findings further reveal common age profiles across cohorts until the age of 60. However, the intercepts differ: in Germany, earlier cohorts are less entrepreneurial than later ones, while in the US, earlier cohorts are more entrepreneurial, with both countries showing an uptick among cohorts born in the 1990s. In the UK, cohort variation is less pronounced; the earliest cohorts show lower entrepreneurship rates than those born from the 1940s to the 1990s, with a decline again for the most recent observable cohort. Year effects are generally less influential than age or cohort effects, though they remain economically meaningful.

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Appendix A Separating age effects from cohort and year effects

To test for the separability of age a and cohort c effects, we follow [Fitzenberger et al. \(2021\)](#) and estimate the following general model, which includes interactions in the age and cohort effects (MODEL 1).

$$y_{ct} = G + \gamma_2 c^2 + \dots + \gamma_p c^p + \alpha_1 a_{ct} + \alpha_2 a_{ct}^2 + \alpha_3 a_{ct}^3 + \sum_i^4 \varphi_i R_i + D^t \beta + u_{ct}.$$

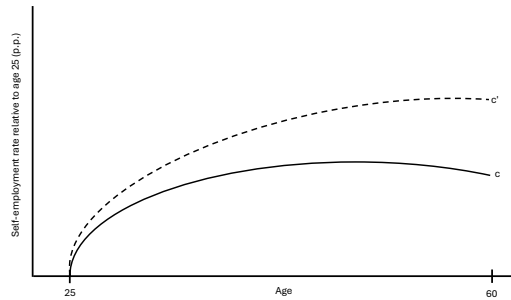
MODEL 1

The term $\sum_i^4 \varphi_i R_i$ consists of the following interaction terms between age a and cohorts c :

$$\begin{aligned} R_1 &= \int a(c + a) da = ca^2/2 + a^3/3 \\ R_2 &= \int a(c + a)^2 da = c^2 a^2/2 + 2ca^3/3 + a^4/4 \\ R_3 &= \int a^2(c + a) da = ca^3/3 + a^4/4 \\ R_4 &= \int a^2(c + a)^2 da = c^2 a^3/3 + 2ca^4/4 + a^5/5. \end{aligned}$$

An interaction of age and cohort effects leads to heterogeneous age profiles between cohorts. A stylized version of these heterogeneous age profiles is depicted in [Figure A.1](#) with cohort c' having a steeper age profile than cohort c .

Figure A.1: Stylized violation: Separability assumption



If age and cohort effects were perfectly separable, all interaction effects would be zero. Therefore, we test for the joint significance of all interaction term coefficients, *i.e.*,

$$H_0 : \varphi_1 = \varphi_2 = \varphi_3 = \varphi_4 = 0. \quad \text{TEST 1}$$

The results are given in the first column of Table A.1. The null hypothesis cannot be rejected for all three countries. Thus, we conclude that interactions between age and cohort effects can be neglected and that the separability condition applies for our sample. Consequently, the slope of the age profile can be considered identical across cohorts.

Table A.1: Hypothesis tests on separability

	Model 1		Model 2
	Test 1	Test 2	Test 3
	$H_0 : \rho_i = 0$	$H_0 : \rho_i = \gamma_i = 0$	$H_0 : \gamma_i = 0$
Germany	0.068	0.000***	0.000***
UK	0.133	0.000***	0.000***
US	0.094	0.000***	0.000***

Note: Table A.1 displays the p-values associated with the specification tests. Test 1 corresponds to the test for the joint significance of the coefficient estimates associated with the interaction terms of the age and cohort effects. Test 2 tests for the joint significance of the coefficients on the interaction terms and cohort terms. Test 3 tests for the joint significance of the coefficient estimates associated with the cohort terms. The p-values are based on Wald-tests, based on a variance-covariance matrix that is clustered on the cohort level. *** p<.001, ** p<.01, * p<.05.

Next, we verify whether the *levels* of the age profiles are identical across cohorts, *i.e.*, whether the age curves share the same intercept across cohorts. Table A.1, column 2, indicates that the joint null hypothesis of the interactions (φ_i) and cohort effects (γ_i) being simultaneously zero in MODEL 1 is rejected. Therefore, we set up a more parsimonious model where we eliminate the interaction terms between age a and cohorts c but retain the cohort polynomials (MODEL 2).

$$y_{ct} = G + \gamma_2 c^2 + \dots + \gamma_p c^p + \alpha_1 a_{ct} + \alpha_2 a_{ct}^2 + \alpha_3 a_{ct}^3 + D^t \beta + u_{ct}. \quad \text{MODEL 2}$$

Again, we test whether the coefficients from the cohort terms in the restricted model are jointly zero, *i.e.*,

$$H_0 : \gamma_1 = \gamma_2 = \dots = \gamma_p = 0. \quad \text{TEST 3}$$

We find that the null hypothesis can be rejected at the .001-percent level (Table [A.1](#), column 3), *i.e.*, significant differences exist in the *level* of the share of self-employment persons across cohorts (*cohort effects*). Hence, the common age profile is identified up to the level of the share of self-employed persons across cohorts.

Based on these findings, we use MODEL 2 for our main specification and estimate the age effects relative to the baseline age. As we consider all individuals between the age of 25 and 60 observed between 1992 and 2019, the baseline age is $a_0 = 25$ and the baseline cohort c_0 comprises all persons born in the year 1969 ($c_0 = t_0 - a_0 = 1992 - 25 = 1967$). Following [Fitzenberger et al. \(2021\)](#), we standardize our variables a and t by subtracting the baseline value and dividing by 10, obtaining $a_{25} = 0$, $a_{26} = 0.1$, $t_{1992} = 0$, $t_{1993} = .1$ and, accordingly, $c_{1967} = 0$, $c_{1968} = 0.1$ etc.

Appendix B Tables

Table B.1: Determinants of the age profile of entrepreneurship

Dimension	Age Pattern	Implications for Entrepreneurship	Sources
Job-specific human capital	Increases with age, peaks in mid-life	Greater ability to identify market niches, understand competitors, and generate business ideas; later in life, skills may become outdated	Schultz (1961); Becker (1991); Mincer (1993); Marvel et al. (2016); Gielnik et al. (2012); Azoulay et al. (2020)
General human capital (fluid vs. crystallized intelligence)	Fluid intelligence peaks in mid-life and declines; crystallized intelligence rises steadily but flattens with age	Mid-life: optimal combination of reasoning and accumulated knowledge; older age: decline in adaptability despite strong knowledge base	Cattell (1963); Kanfer and Ackerman (2004); Salthouse (2009)
Social capital (networks)	Expands with age, peaks in mid-life, declines in older age	Networks mobilize resources (finance, information, customers); mid-life entrepreneurs benefit most from broad networks	Bosma et al. (2004); Aldrich and Zimmer (1986a); Granovetter (1973); Davidsson and Honig (2003a); Stuart and Sorenson (2005)
Financial capital (savings, bequests, access to loans)	Savings follow inverse U-shape; bequests peak in second half of life; loan access declines in older age	Financial resources most abundant in mid-life; younger individuals credit-constrained; older individuals face shorter time horizons	Modigliani and Brumberg (1954); Ando and Modigliani (1963); Carroll (1997); DeNardi et al. (2010); Banks et al. (2004); Krueger (2003)
Access to external finance	Depends on experience and expected working horizon	Mid-life individuals are most attractive to creditors and investors; older entrepreneurs disadvantaged due to shorter time horizons	Azoulay et al. (2020); Stuart and Sorenson (2005)

Table B.2: Cohort moderators of the age profile of entrepreneurship

Dimension	Cohort-Specific Moderators	Illustrative Sources
Job-specific human capital	Declining job stability and shorter tenure spells in advanced economies have shifted incentives away from accumulating firm-specific human capital toward more transferable skills. These changes affect both the baseline level and the slope of the age-entrepreneurship profile.	Goulart and Oesch (2024); Bussolo et al. (2022); Molloy et al. (2024)
General human capital	Rising educational attainment and the expansion of tertiary education have increased general human capital across successive cohorts. While additional schooling may delay entrepreneurship, it raises the cognitive skill base and overall propensity to engage in entrepreneurial activities.	Lee and Lee (2016); Schofer and Meyer (2005); Carlsson et al. (2015); Ahn and Winters (2023); Fritsch et al. (2015)
Social capital	Changes in communication technologies, especially the rise of social media platforms, have transformed how networks are built and maintained. These developments may generate both level shifts and slope changes in the social capital-entrepreneurship profile across cohorts.	Putnam (2000); Ellison et al. (2007); Burt (2005)
Financial capital	Structural economic shifts from manufacturing toward services have lowered capital requirements for entry, while financial liberalization has expanded access to external resources. Both forces may raise baseline entrepreneurship rates and alter the age trajectory across cohorts.	Fourastié (1949); Baumol (1967); Schettkat and Yocarini (2003); Herrendorf et al. (2014); Autor (2015); Black and Gilson (1998); Greenwood and Scharfstein (2013)

Table B.3: Regression results

	Germany		UK		US	
Dependent variable: self-employment rate in the whole population						
<i>age</i>	0.062	***	0.052	***	0.063	***
	(0.004)		(0.006)		(0.007)	
<i>age</i> ²	-0.013	***	-0.012	***	-0.025	***
	(0.003)		(0.003)		(0.005)	
<i>age</i> ³	0.001		0.001		0.003	***
	(0.001)		(0.001)		(0.001)	
<i>cohort</i> ²	-0.007	***	-0.002	*	0.001	
	(0.001)		(0.001)		(0.001)	
<i>cohort</i> ³	0.002	***	-0.003	***	-0.003	***
	(0.000)		(0.001)		(0.001)	
<i>cohort</i> ⁴	0.001	***	0.000		0.000	
	(0.000)		(0.000)		(0.000)	
<i>cohort</i> ⁵	-0.000		0.000	***	0.000	***
	(0.000)		(0.000)		(0.000)	
Intercept	0.027	***	0.069	***	0.042	***
	(0.003)		(0.005)		(0.004)	
Time effects	yes		yes		yes	
Number of observations	1008		1008		612	
Adjusted <i>R</i> ²	0.71		0.66		0.55	

Note: *** $p < .001$, ** $p < .01$, * $p < .05$.

Table B.3 lists the regression results from estimating Equation 1. The dependent variable is the share of self-employed persons in the whole population. The level of observation is cohort c in year t in country i , *e.g.*, the share of self-employed persons among all persons born in 1969 living in Germany in the year 2000, such that $N_i = C_i \times T_i$. In total, we observe 36 cohorts over 28 years in Germany and in the UK and over 17 years in the US, since the PSID surveyed households biennially after 1998.

Table B.4: Hypothesis tests on separability for age range 25 to 75

	Model 1		Model 2
	Test 1	Test 2	Test 3
	$H_0 : \rho_i = 0$	$H_0 : \rho_i = \gamma_i = 0$	$H_0 : \gamma_i = 0$
Germany	0.000***	0.000***	0.000***
UK	0.005**	0.000***	0.000***
US	0.000***	0.000***	0.000***

Note: Table B.4 displays the p-values associated with the specification tests on separability for a sample of 25- to 75-year-olds. Test 1 corresponds to the test for the joint significance of the coefficient estimates associated with the interaction terms of the age and cohort effects. Test 2 tests for the joint significance of the coefficients on the interaction terms and cohort terms. Test 3 tests for the joint significance of the coefficient estimates associated with the cohort terms. The p-values are based on Wald-tests, based on a variance-covariance matrix that is clustered on the cohort level. *** $p < .001$, ** $p < .01$, * $p < .05$.

Table B.5: Tests on inverted U-shape for different age ranges

	steepness at the tails		turning points	
	left tail	right tail	a_{tp} s.t.	
	$H_0 : f'(a_L) = 0$	$H_0 : f'(a_R) = 0$	$f'(a_{tp}) = 0$	
<i>between 25 and 70 years</i>				
Germany	0.000***	0.000***	50.3	
UK	0.000***	0.000***	47.2	2.8
US	0.000***	0.371	51.0	78.1
<i>between 25 and 75 years</i>				
Germany	0.000***	0.000***	53.8	120.2
UK	0.000***	0.000***	50.5	−45.9
US	0.000***	0.000***	46.4	

Note: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Table B.5 reports the results for the inverted U-shape tests following Lind and Mehlum (2010) considering extended age ranges up to age 70 and 75, respectively. The first column reports the p-value from testing the null hypothesis that the first derivative of the age polynomial is zero at age 26. The second column reports the p-value from testing the null hypothesis that the first derivative of the age polynomial is zero at age 70 and 75, respectively. The third and fourth columns report the turning points of the estimated age polynomial, *i.e.*, the age level for which the first derivative of the age polynomial is equal to zero. The separability condition in Germany is not fulfilled for age range 25 to 70 and 25 to 75, *i.e.*, there is no common age profile across cohorts. Neither the US nor the UK fulfills the separability condition for the age range 25 to 75.